Daniel Marta

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Research Statement

I am a postdoctoral researcher at ETH Zürich working on world models, vision-language-action (VLA) models, uncertainty-aware policy learning with large vision-language model (VLM) backbones, reinforcement learning, and large language models. I envision a world where robots and artificial agents interact safely and adapt seamlessly to humans through diverse feedback while knowing when their actions may be unreliable. My research focuses on aligning autonomous systems with human and environmental feedback: preference-based reinforcement learning for safe and sample-efficient robot adaptation [C2, C4, C5, C7], dense credit assignment from rich textual feedback for efficient LLM post-training [C6], and uncertainty quantification for failure-aware and sample-efficient VLA adaptation [W1].

Keywords: World Models, Vision-Language-Action (VLA) Models, Uncertainty Quantification, Vision-Language Models (VLMs), Reinforcement Learning (RL), Large Language Models (LLMs), RL from human feedback (RLHF)

Selected Publications

C=Conference, J=Journal, W=Workshop, T=Thesis, *=equal contribution

[W1] Römer, R., Seeliger, M., Liu, S., Sturgis, B., Bagatella, M., **Marta, D.**, Krause, A., & Schoellig, A. P. (2026). Uncertainty Quantification for Flow-Based Vision-Language-Action Models. Accepted to the RSS 2026 Diff4RL Workshop. arXiv preprint arXiv:2606.18043.

[C6] Hübotter, J., Lübeck, F., Behric, L. D., Baumann, A., Bagatella, M., **Marta, D.**, Hakimi, I., Shenfeld, I., Kleine Buening, T., Guestrin, C., & Krause, A. (2026). Reinforcement Learning via Self-Distillation. In International Conference on Machine Learning (ICML). Also accepted to the ICLR Workshop on Scaling Post-training for LLMs (SPOT).

[C7] **Marta, D.***, Holk, S.*, & Leite, I. (2026). MOSAIC: Multi-objective optimization from zero-shot language reasoning in preference-based RL. In 2026 IEEE International Conference on Robotics and Automation (ICRA).

[C5] **Marta, D.***, Holk, S.*, Vasco, M., Lundell, J., Homberger, T., Busch, F., Andersson, O., Kragic, D., & Leite, I. (2025). FLoRA: Sample-Efficient Preference-based RL via Low-Rank Style Adaptation of Reward Functions. In 2025 IEEE International Conference on Robotics and Automation (ICRA).

[C2] Holk, S., **Marta, D.**, Pek, C., & Leite, I. (2024, May). POLITE: Preferences combined with highlights in reinforcement learning. In 2024 IEEE International Conference on Robotics and Automation (ICRA) (pp. 2288–2295). IEEE. **Finalist for the ICRA 2024 Best Conference Paper Award, Best Student Paper Award, and Best Paper Award on Human-Robot Interaction.**

[C4] **Marta, D.***, Holk, S.*, Pek, C., & Leite, I. (2024, May). SEQUEL: Semi-Supervised Preference-based RL with Query Synthesis via Latent Interpolation. In 2024 IEEE International Conference on Robotics and Automation (ICRA) (pp. 9585-9592). IEEE.

[W2] Yang, Y., Zhang, Q., Li, C., **Marta, D.**, Batool, N., & Folkesson, J. (2024). Human-Centric Autonomous Systems With LLMs for User Command Reasoning. In 2024 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW), LLVM-AD Workshop (pp. 988–994). **Best Student Paper Award.**

Experience

- **Postdoctoral Researcher** — *2025 - Present*
  ETH Zürich — Zürich, Switzerland
  - Conducting research on world models and vision-language-action (VLA) models with large vision-language model (VLM) backbones.
  - Developing reinforcement learning and large language model methods for robotic and agentic systems.
- **Visiting Researcher** — *2024 - 2024*
  Interactive Machines Group, Computer Science Department, Yale University — United States
  - Explored enhancing interpretability of human feedback when modeling reward functions in the context of RLHF.
- **Research Engineer** — *2018 - 2018*
  RL Team, Waymo Research — United Kingdom
  - Implemented distributed RL agent learning in Docker containers for self-driving car simulations.
- **Research Intern** — *2017 - 2018*
  RL Team, Waymo Research — United Kingdom
  - Implemented RL algorithms such as DDPG and DQN.
- **Graduate Research Fellow** — *2016 - 2017*
  GAIPS, INESC-ID — Portugal
  - Researched unsupervised learning methods within RL contexts.

Education

- **PhD in Computer Science** *2020 - 2025*
  RPL (Robotics, Perception and Learning) Department, KTH Royal Institute of Technology Stockholm, Sweden
  - Completed PhD in 2025.
  - Enhanced Preference-based RL methods for robot control tasks which can adapt to human feedback.
  - Devised RLHF methods to enhance query information for human-robot interaction.
  - Supervisors: Prof. Iolanda Leite, Prof. Jana Tumova.
- **Doctoral Researcher** *2019 - 2020*
  ATO (Air Transport and Operations) Department Delft, Netherlands
  - Investigated combinatorial optimization in RL to predict aircraft scheduling and passenger preferences for air transport operations.
- **B.Sc. & M.Sc. in Aerospace Engineering, Avionics (Grade: 16/20)** *2009 - 2016*
  Technical University of Lisbon Lisbon, Portugal
  - Thesis: *Deep Learning Methods for Reinforcement Learning*.
  - Supervisor: Prof. Rodrigo Ventura.

Honors and Awards

- **Best Paper Awards**
  - **Best Student Paper Award** for an outstanding contribution to the WACV 2024 LLVM-AD workshop.
- **Nominations**
  - POLITE was a finalist for the ICRA 2024 **Best Conference Paper Award**, **Best Student Paper Award**, and **Best Paper Award on Human-Robot Interaction**.
- **Grants**
  - Awarded a **prestigious international postdoctoral fellowship** from the **Knut and Alice Wallenberg Foundation**.
  - Awarded research visits: **Yale University** and WASP Japan **AI manufacturing** research visit, visiting Toyota, Toyota Industries, Toshiba, Hitachi, and DMG Mori.
  - Awarded a **Wallenberg-funded WASP PhD Fellowship/Grant**.

Teaching and Supervision

- **Teaching** *2020 - 2024*
  DD2380 AI course
  - Teaching assistant and leader of the RL assignment; lectured RL tutorials and coordinated teams of teaching assistants to deliver lectures and in-person examinations.
  - Developed and maintained the RL assignment and the dockerized evaluation process.
- **Supervision**
  - Oskar Stigland, Year:2024, Thesis: Explainability and Hierarchical Reinforcement Learning.
  - Mohammed Akif, Year:2023, Thesis: Safe Reinforcement Learning for Social Human-Robot Interaction.

Skills

- **Agentic AI Systems:** Advanced operation of self-hosted agents: tool integrations, cron/sub-agents, Telegram, and Linux/VPS deployments; upstream Hermes Agent (PR #22823) and OpenClaw (PR #49315) contributor.
- **LLMs & VLMs:** Proficient with open-source LLMs/VLMs used in SDPO: Llama, Mistral/Mixtral, DeepSeek, GLM, and Qwen examples (Qwen2.5/Qwen3/Qwen3-Coder; Qwen2.5-VL/Qwen3-VL).
- **Programming & Scripting Languages:** Proficient in Python, including socket networking and multi-processing/threading; moderate experience with C++, Javascript, SQL, XML/XSL, LaTeX; minor experience with Rust.
- **Machine Learning Libraries:** Proficient in PyTorch, TensorFlow, and Hugging Face Transformers/Datasets; contributed to RL tooling in OpenAI Baselines (PR #158) and rllab (PR #186).
- **Simulation Engines:** Implemented several RL environments in Unity, familiar with Python MCPs (Model Context Protocols), moderate experience in Unreal Engine.
- **Containerization & Deployment:** Proficient in Docker, Linux/VPS operations, SSH, services, reverse tunnels, remote workstations/HPC workflows; moderate Kubernetes.
- **Additional Tools:** Moderate experience with Angular and SolidWorks; minor experience with Blender and Photoshop.

Academic Service

Reviewer: IEEE International Conference on Robotics and Automation (ICRA, 2020–2024); IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS, 2022–2023); IEEE Robotics and Automation Letters (2022–2023); ACM/IEEE International Conference on Human-Robot Interaction (HRI, 2023–2024).